

Crop Tracking Project — Operations & Analytics Dashboard Architecture

This specification details the layout, data metrics, visual components, and real-time interface design for the **Crop Tracking System Admin & Analytics Dashboard**.

1. Key Performance Indicators (KPIs) Summary

The dashboard provides real-time visibility into overall farm health, device connectivity, and active field operations.

Metric Name	Current Value	Target / Status	Trend
Total Active Fields	18 Zones	Optimal Monitoring	+2 this month
Overall Crop Health Score	92.4%	Target > 90%	+1.8% vs last week
Active Field Devices / Sync Status	14 / 15 Online		
Pest / Disease Detection Rate	3 Alerts Active	Low Severity (Early Blight)	-2 alerts from yesterday
Projected Yield (Current Cycle)	42.5 Tons	Estimated Harvest: 14 Days	+5% vs previous cycle

2. Dashboard Layout & Widget Breakdown

2.1 Field Health & Disease Map View

- **Interactive GIS Field Map:** Visual color-coded overlay (Green = Healthy, Yellow = Attention Needed, Red = Disease Alert) based on mobile image scan submissions.
- **Quick Filter Bar:** Filter by crop type (e.g., Maize, Tomatoes, Beans), growth phase, or field location.

2.2 Sync Queue & Offline Device Monitor

- **Device Health Panel:** Real-time status of connected mobile app clients and field sync queues.
- **Pending Mutation Tracker:** List of un-synced field observations and image uploads awaiting server verification.

2.3 Activity & Fertilizer Tracking Feed

- **Live Activity Stream:** Chronological feed of field worker logs (irrigation completed, pesticide applied, crop scan uploaded).
- **Task Scheduling Widget:** Upcoming automated reminders for field maintenance and harvest preparation.

3. Dashboard API Data Model Example

```
{
  "dashboard_summary": {
    "total_fields": 18,
    "active_crops": ["Maize", "Tomatoes", "Cassava"],
    "health_index": 0.924,
    "active_alerts": [
      {
        "alert_id": "ALT-102",
        "field_name": "North Field - Plot B",
        "issue": "Early Blight detected via ML Scan",
        "severity": "MEDIUM",
        "timestamp": "2026-10-08T00:15:00Z"
      }
    ],
    "sync_stats": {
      "active_mobile_users": 15,
      "pending_sync_records": 4
    }
  }
}
```

```
}  
}  
}
```

4. Implementation Roadmap for Dashboard Development

1. **Phase 1: UI Mockups & Component Library Setup** — Build reusable React dashboard components, summary cards, and layout grids.
2. **Phase 2: WebSocket & Live Data Integration** — Connect UI components to backend REST and WebSocket APIs for live sync and activity feeds.
3. **Phase 3: Interactive Maps & Charting** — Implement GIS mapping layer for field boundaries and visual analytics for yield trends.
4. **Phase 4: User Roles & Export Options** — Implement role-based access for farm managers and export functions (PDF/CSV yield reports).